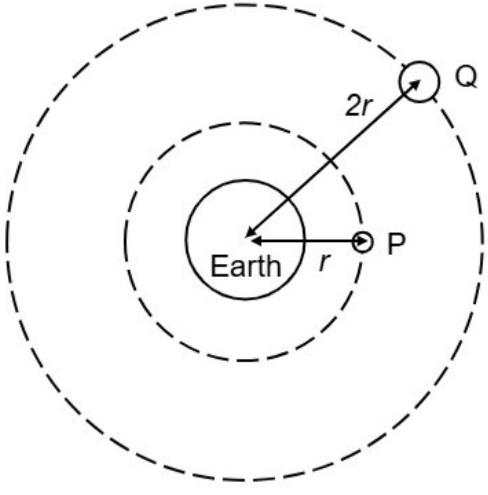
8 Two satellites P and Q are in circular orbits about the Earth. Satellite P has mass m and it is in an orbit of radius r. Satellite Q has mass $2m$ and it is in an orbit of radius $2r$.



Which of the following statements is true?

A The speed of satellite Q is greater than that of satellite P.

B The speed of satellite Q is the same as that of satellite P.

C The kinetic energy of satellite Q is greater than that of satellite P.

D The kinetic energy of satellite Q is same as that of satellite P.